

# JSPL EAF Digital Twin Architecture Roadmap

Phase 27 recommendation document — experimental research track. Production remains frozen.

## 1. Version ladder

### V1 — Current planning model

Frozen Phase 19 StackingRegressor (leaky EE-derived features) serves production. Experimental Phase 25 two-stage + Phase 26 gold leakage-free features ready for shadow mode only.

### V2 — Planning + SCADA/MES features

Ingest P0 tags: power-on/off, delay codes, charging timestamps, restriction flag; P1: metallization, HM temp. Retrain leakage-free Stage 2; improve Stage 1 delay classifier.

### V3 — Real-time digital twin

Historian streams feed live TTT residual prediction, foam proxies from V/I, and delay early-warning. Operator dashboard shows confidence and SHAP drivers.

### V4 — Closed-loop recipe recommendation

Optimizer (Phase 20 lineage) constrained by twin forecasts, EMS limits, and metallurgical rules — human confirm in the loop.

## 2. Data flow

**PLC / Instrumentation** → **SCADA** (electrical, gas, electrodes) → **Historian** → **MES** (heats, delays, charge plans, quality) → **Feature store** → **AI modules** (regime classifier, TTT regression, delay warner) → **Operator dashboard & Optimization engine** → confirmed setpoints back to MES/SCADA.

## 3. AI modules

| Module                | Role               | Depends on                            |
|-----------------------|--------------------|---------------------------------------|
| Regime classifier     | NORMAL vs ABNORMAL | Delay codes, power-off, restriction   |
| Normal TTT regressor  | Planning estimate  | Recipe + power-on drivers + quality   |
| Delay warner          | Early alert        | Event streams + logistics delays      |
| Energy / foam advisor | Process coaching   | V/I, C/O2 profiles                    |
| Optimizer             | Recipe search      | Twin constraints, not EE_KWH free var |

See [presentation\\_figures/digital\\_twin\\_layers.png](#) for the maturity diagram.